DeepStateNet: A Deep Learning-Based Multimodal EEG Classifier for Motor Imagery Using Microstate Dynamics

MASTER PROJECT
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> *"Ang hindi marunong lumingon sa pinanggalingan ay hindi makararating sa paroroonan." ("He who does not look back to where he came from will never reach his destination.")*
>
> (Filipino proverb)

Contents

1 Introduction

1.1 Brain–Computer Interface (BCI)

Brain Computer Interfaces (BCIs) represent a rapidly developing interdisciplinary field that integrates neuroscience, engineering, and computer science to establish direct communication pathways between the brain and external devices. Initially conceptualized in the 1970s, BCIs have evolved from proof-of-concept demonstrations (e.g., cursor movement) to sophisticated tools capable of controlling multi-degree-of-freedom robotic prostheses, supporting neurorehabilitation, and augmenting human capabilities [1–4]. Modern BCI technology leverages advances in machine learning, signal processing, and neuroscientific understanding to decode patterns of neural activity into actionable commands [5–8].

1.1.1 Electroencephalogram (EEG) based BCI

Among various neural recording, Electroencephalography (EEG) is one of the most used in non-invasive BCIs due to its safety, portability, cost-effectiveness, and high temporal resolution [1–3, 8, 9]. EEG measures voltage fluctuations on the scalp arising from synchronized postsynaptic potentials of large neuronal populations [10, 11]. High-density EEG provides rich spatial and temporal datasets, enabling the study of large-scale brain network dynamics relevant for both cognitive and motor tasks [10, 12]. EEG-based BCIs often operate either with *endogenous signals* (e.g., sensorimotor rhythms modulated during Motor Imagery) or *exogenous signals* (e.g., P300, steady-state visual evoked potentials) [1, 8].

1.1.2 Utility of MI-BCI

Motor Imagery (MI) the mental simulation of motor actions without overt execution modulates sensorimotor rhythms (α: 8–12 Hz; β: 18–26 Hz) over the contra-lateral cortex [4, 8, 13]. MI-based BCIs (MI-BCIs) translate these EEG rhythms into real-time control signals for devices such as robotic arms, wheelchairs, spellers, or rehabilitation robots [8, 9, 14, 15]. MI has been shown to activate similar cortical patterns as actual movements and is a potential facilitator of neuroplasticity. MI-BCIs are especially relevant for stroke, amyotrophic lateral sclerosis (ALS), spinal cord injury (SCI), Parkinson's disease, and multiple sclerosis rehabilitation due to their ability to induce neuroplasticity and engage motor cortical areas even in patients with severe paralysis [1, 2, 16, 17]. Visually guided MI (GMI) or kinesthetic MI paradigms may enhance cortical activation and improve motor recovery outcomes [8, 17]. However, inter-subject variability (BCI illiteracy) remains a major challenge, with about 20% of users unable to gain effective control. Predictive approaches using resting-state EEG features such as spectral entropy or microstate metrics are being developed to screen and adapt systems to user capabilities [9].

1.1.3 Machine Learning

Classical EEG decoding pipelines typically involve preprocessing, feature extraction, and classification [1, 8]. Popular algorithms include Common Spatial Patterns (CSP) and Filter-Bank CSP (FBCSP), which extract spatial filters maximizing variance differences between classes across frequency bands [6, 15]. Spatial features are then classified using algorithms such as Linear Discriminant Analysis (LDA) [8, 16], Support Vector Machines (SVMs) [17, 18], k-Nearest Neighbors (kNN), Naïve Bayes, or Decision Trees [8, 19]. Regression-based decoders like Partial Least Squares (PLS) [20], Kalman filters, or Ridge regression are used for continuous outputs [8]. While effective for individual-specific models, these methods often fail to generalize across subjects due to high EEG variability [6, 9]. There is also the limitations of traditional machine learning (ML), which relies on predefined, subject specific features that are often limited.

1.1.4 Deep Learning

Deep learning (DL) approaches bypass handcrafted feature engineering by automatically learning hierarchical spatial, spectral and temporal representations from raw EEG [5, 6, 13, 19]. Convolutional Neural Networks (CNNs) such as DeepConvNet, ShallowConvNet [5], EEGNet [13], and their hybrids with LSTMs or Transformers have demonstrated competitive or superior performance in MI decoding tasks [2, 3, 6, 19]. CNNs learn spatial and spectral filters, while Recurrent Neural Networks (RNNs) and LSTMs capture temporal dependencies. Transformers and attention-based networks can model long-sequence dependencies and dynamically weight important EEG segments [3]. Transfer learning and domain adaptation address inter-subject variability by adapting pre-trained models to new users [6]. Despite high accuracy, deep networks require substantial data and risk overfitting, motivating EEG-specific architectures with fewer parameters, such as EEGNet [13].

1.2 Microstates

1.2.1 Definition

EEG microstates are series of quasi-stable states with particular topographical patterns of electric potentials that can last for approximately 30–120 ms before abruptly changing to another state [9–11, 20, 21]. First theorized in the 1970s, microstates have since become a well-established concept, particularly within cognitive research [21–23]. They reflect coordinated activity of large-scale brain networks and have been described as the "atoms of thought" the basic building blocks of cognition [12]. Microstate analysis models the EEG as a sequence of these discrete, recurring maps, capturing whole-brain temporal dynamics at the millisecond scale [9, 10].

1.2.2 State of the Art

Resting-state EEG typically reveals four canonical microstates (A–D) explaining nearly 70% of the global variance, each associated with known functional networks (e.g., auditory, visual, salience, attention) [10, 21]. However, studies show that motor tasks including MI induce new task-specific microstates while retaining some resting-state patterns [24]. Microstate parameters mean duration, occurrence rate, time coverage, transition probabilities are sensitive to cognitive states, personality traits, neuropsychiatric conditions, and BCI aptitude [9, 20, 21]. Notably, other studies have labeled microstates differently (MS1–MS4), reporting, for instance, longer durations of MS3 and a reduced occurrence of MS1, both correlating positively with MI-BCI performance [9]. More recently, a 2024 study suggested increasing the number of canonical microstates to five, reflecting a growing body of work reporting higher microstate counts [23].

1.2.3 Existing Models

The dominant method for microstate segmentation applies modified k-means clustering (or hierarchical clustering) to EEG maps at peaks of Global Field Power (GFP). Optimal cluster number is determined via cross-validation, Krzanowski–Lai indices, or permutation testing [9–12]. The resulting template maps are back-fitted to the continuous EEG to yield symbolic microstate sequences. Toolkits like CARTOOL, EEGLAB plugins, and Pycrostates implement these workflows [12, 25, 26], with options for subject-level or group-level templates, smoothing constraints [11], and the integration of advanced metrics [24]. Variants of the model incorporate independent component analysis (ICA) and fractal/scale-free temporal dynamics analysis, linking them to metastable network switching relevant for flexible cognition and BCI control [10].

The Global Field Power is calculated with the following formula:

$$\mathrm{GFP} = \sqrt{\frac{\sum_{i=1}^{N} (u_i - \bar{u})^2}{N}} \tag{1}$$

where u_i is the voltage of the map u at the electrode i, $\bar{u}$ is the average voltage of all electrodes of the map u and N is the number of electrodes of the map u.
Although most models have been applied to clustering, a few have explored the use of microstates as features. For example, Cui et al. (2023) developed a MI-BCI predictor relying exclusively on microstate features such as mean duration, occurrence rate, time coverage ratio, and transition probability [9]. Their regression model obtained a relatively high performance, though it was applied specifically in the context of bilateral prediction.

1.3 Motivation

As mentioned previously, both MI-BCI and EEG microstates have been introduced as well established and largely independent domains. While there is some application of machine learning to microstate analysis, these remain relatively limited and rarely considered microstates directly as features. This paradigm is still largely unexplored.
This project aim to deliberately incorporate microstate features into a deep learning model for motor imagery classification. The purpose is to bring neurological insight into the data-driven pipeline, even if it comes at the cost of breaking with the end-to-end principle of deep learning. The main purpose is to research whether such structured prior knowledge can provide complementary information and improve the interpretability of BCI models.

2 Materials & Methods

2.1 Dataset

The dataset analyzed in this study originates from recordings collected for the development of the CastNet model, which investigated neural activity associated with right-hand open and close motor attempt tasks [3]. Data acquisition was carried out at the Centre for Brain-Computing Research (CBCR), Nanyang Technological University NTU, Singapore, with approval from the Institutional Review Board (IRB-2018-06-030).

2.1.1 Participants and Electrophysiological Recording

Fifty healthy adults participated in the recordings (31 males, 19 females; mean age 27.04±5.15 years). In order to assess the dominant hand and confirm that participants are indeed right handed, the Edinburgh Handedness Inventory-Short Form was given to the participants. All EEG were recorded with a BrainAmp ActiCHamp amplifier using 61 electrodes positioned according to the international 10–20 system, referenced to FCz, at a sampling rate of 1000 Hz.

2.1.2 Experimental Design and Procedure

Participants were seated in front of a computer display and fitted with both the EEG cap and a Handy Rehab^{TM}, an intelligent robotic glove for rehabilitation on the right hand. Visual cues, delivered through a custom graphical user interface (GUI), instructed participants to attempt either hand opening or hand closing actions. This restricted the gaze direction. The glove standardized the preparation phase by actively positioning the hand before each trial. Each session comprised ten blocks of 20 trials (10 *Open*, 10 *Close*), resulting in 200 trials per subject. Breaks were provided between blocks to mitigate fatigue.
Each trial consisted of three phases:

- **Baseline (2 s):** A fixation cross appeared on the screen.
- **Preparation ($\leqslant$ 6s):** The robotic glove moved the hand into the required state (open or closed). If the hand was already in the correct position, the phase was shortened to 4 s, corresponding to the *Rest* condition.
- **Action (4 s):** A text cue instructed the participant to attempt the required movement while minimizing overt muscular contractions.

2.2 Preprocessing and Dataset Structure

The dataset was delivered in an already preprocessed form. The reported preprocessing steps included band-pass filtering between 0.3 to 40 Hz (zero-phase Chebyshev Type II IIR filter), then a Common Average Referencing, and Independent Component Analysis (FastICA, implemented in MNE-Python) for attenuation of ocular and muscular artifacts. Finally, any remaining artifacts observed during a visual inspection were removed.

To ensure balanced class distribution between *rest* and their respective action trials (*open* or *close*, all *rest*, and an equal number of *Open* and *Close* trials were selected (maximum 100 per class; otherwise, the first N available trials were kept). On average, this procedure resulted in approximately 360 artifact-free trials per subject in the final dataset containing both open and close trials and there respective rest trials (see tab. 1). Giving an unbalanced dataset of 50% *rest*: class 0, 25% *open*: class 1 and 25% *close*: class 2.

Table 1: Trial Count Summary Statistics by Condition (N=50 subjects)

Condition	Mean	Std Dev	Min	Max	Range
Rest	179.2	14.4	140.0	200.0	60.0
Open	89.6	7.2	70.0	100.0	30.0
Close	89.6	7.2	70.0	100.0	30.0
Total	358.4	28.8	280	400	120

Note: Rest condition has twice as many trials as Open/Close conditions by design. Open and Close conditions have identical statistics due to balanced experimental design. Rest is obtain from both Open and Close experiment prior each action block

2.3 Case Studies

In Han Wei Ng. et al's (2024) study from which the datasets comes, three different case study, *dependent subject*, *independent subjects* and *adaptive subjects* were explored [3]. This project focused on dependent subject due to the variability of microstates between subjects.

Subject-dependent: Each model is specially designed for one subject, using only their data. Each goes through a five-fold cross-validation paradigm to train and validate the models. This method consist of splitting the dataset of the subject into 5 equally distributed and same fold size. Each fold is used as a validation set while the remaining four will be the training set. This repeats for each fold and the means and standard deviation (std) of the five folds are reported. This method allows to have a more robust assessment compared to a regular split. It ensures that one split is not over or underfitting while keeping a ratio of 80/20% training and validation set.

Due to the nature of the dataset with an imbalanced in class, the scores used for validation and models quality assessment are *balanced accuracy* and *F1 macro*.

2.4 Microstates extraction

In order to treat microstates and extract features like in the literature. Trials or GFP time-series needs to be segmented into microstates sequences. To generate them many methods exist. One of the most commonly use nowadays is clustering models to select the best clusters at GFP peaks as mentioned in section 1.2. To do this, the best number of cluster, i.e. microstates, must be determined.

2.4.1 Number of clusters selection

The optimal number of clusters in microstates analysis is not fixed and often varies across studies and individuals [21, 23, 24]. In the present work, two complementary strategies were adopted. First, a reference solution of $K = 5$ clusters was employed, following the recent recommendations by Michel et al. (2024) in their review on the current state of EEG/ERP microstate research [23]. This choice reflects the most widely reported canonical microstate classes during resting and steady-state activity.

Second, to adapt the clustering procedure to the present dataset, an exploratory model selection was performed using the *ModKMeans* algorithm implemented in *Pycrostates* [26], based on Pascual-Marqui et al. [11]. For each subject, clustering solutions were computed for $K = 4$ up to $K = 50$ clusters. To evaluate each solution, three quality metrics were extracted: (i) the Global Explained Variance (GEV), (ii) the mean spatial correlation between scalp maps and their assigned centroids, and (iii) the Calinski–Harabasz (CH) index [27].

The GEV was computed as

$$\mathrm{GEV} = \frac{\sum_t \left(\|x_t\|\, r_{t,c(t)}\right)^2}{\sum_t \|x_t\|^2}, \tag{2}$$

where $x_t \in \mathbb{R}^N$ denotes the scalp potential map at time t, $r_{t,c(t)}$ its correlation with the assigned centroid, and $\|\cdot\|$ the Euclidean norm. A higher GEV indicates that the extracted clusters explain a greater proportion of the global EEG variance.

The spatial correlation between a map $x \in \mathbb{R}^N$ and its centroid $y \in \mathbb{R}^N$ was defined as

$$\textit{spatial correlation}(x, y) = \frac{x \cdot y}{\|x\|\,\|y\|}, \tag{3}$$

with absolute values taken to enforce polarity invariance, consistent with the ModKMeans assignment rule.

Finally, the Calinski–Harabasz index quantifies cluster compactness and separation:

$$\mathrm{CH}(K) = \frac{\mathrm{Tr}(B)}{\mathrm{Tr}(W)} \cdot \frac{n-K}{K-1}, \tag{4}$$

where $\mathrm{Tr}(B)$ and $\mathrm{Tr}(W)$ denote the trace of the between- and within-cluster dispersion matrices, respectively, and n is the number of samples. Higher CH values indicate more distinct and compact clustering solutions.

2.5 Models

For this project, two models were build in order to classify the motor intent of the subjects using microstates. *MicroStateNet (MSN)* a microstates-based convolutional Neural Network inspired by *DeepConvNet (DCN)* [5]. And the final model that combine MSN and DCN in a multimodal fashion to extract EEG and Microstate features, *DeepStateNet (DSN)*. Each model's architecture is describe in the following subsection:

2.5.1 DeepConvNet

DeepConvNet was used both as a baseline model for comparison, as inspiration for MicroStateNet and as a component of the DeepStateNet. It has the following architecture:
The model takes EEG input in the form of a two-dimensional array (time × channels). Each block has a 2-dimensional convolution layer to extract both temporal and spatial features. Four succeeding convolution–pooling blocks progressively increase the number of feature maps (approximately 25→50→100→200) while applying temporal convolutions, batch normalization, dropout, and ELU activations. Each block ends with max-pooling (stride 3×1). The final layer is a dense classifier with three softmax outputs corresponding to the hand movement classes [5].
This design allows the initial layers to extract frequency-specific temporal filters and spatial patterns (similar in spirit to CSP), while deeper layers capture higher-level, task-relevant abstractions directly from raw EEG.

Table 2: DeepConvNet architecture based on Schirrmeister et al. [5]. The table summarizes the input, four convolution–pooling blocks, and final classifier, including approximate kernel sizes, filter counts, and pooling strides.

Component	**Temporal kernel size**	**Layers (conv / pooling)**	**Pooling type (stride)**
Input	time × electrodes (EEG channels)	—	—
Conv–Pool Block 1	10	Temporal conv (25 filters) → Spatial conv (25 filters) → Pooling	Max pooling (3×1)
Conv–Pool Block 2	10	Conv (≈50 filters) → Pooling	Max pooling (3×1)
Conv–Pool Block 3	10	Conv (≈100 filters) → Pooling	Max pooling (3×1)
Conv–Pool Block 4	10	Conv (≈200 filters) → Pooling	Max pooling (3×1)
Classifier layer	3 output units	Dense linear layer → Softmax	—

2.5.2 MicroStateNet

MicroStateNet is modified version of DeepConvNet adapted for microstates sequences (see Figure 1). Indeed, due to the shape and nature of the input, a few adaptation were necessary:

- microstates sequences don't have spatial dimension
- microstates sequences are categorical

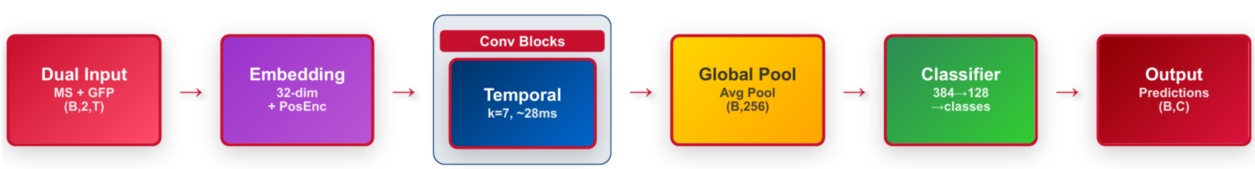

Figure 1: MicroStateNet Architecture with kernel size = 7

Contrary to EEG data microstates sequences don't have channels. Hence, a 2D convolutional layer would not make sens as there is no spatial dimension, which means that a 1D convolutional layer suffice (see Figure 2). Furthermore, since a microstate sequence is categorical, values (microstates) don't have continuous meanings. This makes it incompatible with a CNN, meaning the input needed to be formatted. Two solutions were tested:

- One-Hot encoding
- Embedding

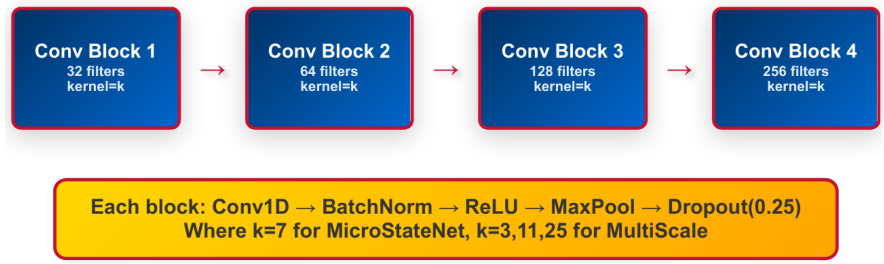

Figure 2: Architecture of a Conv Block for kernel size k

A one-hot encoding was first considered for microstates sequences as it is a well used tool in machine learning when working with categorical features. Plus the shape of the data would mimic, with reduced dimension, the structure of an EEG dataset. This methods consist of enhancing the dataset with n sequences for each categorical timeseries. This means for n microstates, there should be n one-hot sequences, which are sequences of '0's and '1's. 1 meaning that the corresponding microstate was present at that time t and 0 if not. However, CNN might not work well with one-hot encoding due to the loss in relation between microstates. Indeed, by using this method there are distinct sequences for each microstates and each are treated independently.

To overcome this issue, embedding was considered as an another encoding. This method consists of representing an object, word or number in a higher *embedded* dimension. With this, microstates are no longer represented by one number, but as a vector. This enhanced the dimension of the sequences, like the one-hot encoding method, while keeping a connection between each microstates.

In addition, because microstates vary from $30 - 120\mu s$ [21, 23], the initial kernel size of MicroStateNet was 7, with a sampling rate of $250Hz$, this gives roughly $28\mu s$ of window. However, due to microstates large range in durations mentioned in section 1.2, the kernel size parameter had to be explored. Hence, the creation of MicroStateNet with multiscale. This version is designed to capture information across multiple temporal scales simultaneously. It consists of three parallel branches, each with a different kernel size (see Figure 3):

- kernel size = 3, for short, smaller than the average microstate duration ($\simeq$ 12 ms),
- kernel size = 11, for lower and middle range of typical microstate durations ($\simeq$ 44 ms),
- kernel size = 25, covering longer temporal segments of microstates (up to $\simeq$ 100 ms).

Finally, for computational efficiency, the depth of each branch was reduced to three layers instead of four in the multiscale MSN version.

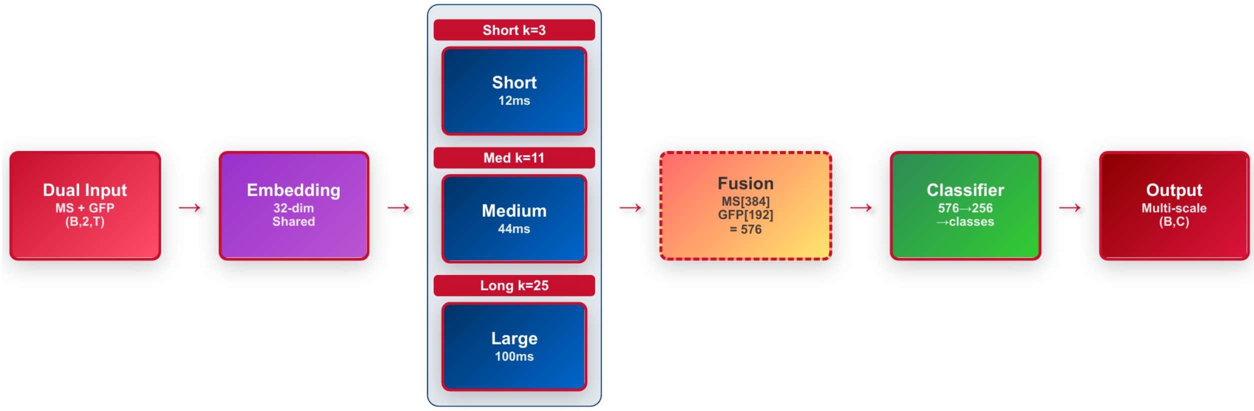

Figure 3: Multiscale MicroStateNet Architecture

2.5.3 DeepStateNet

Realizing that microstates features have their own structure, a new model was necessary to extract them. This meant breaking the end-to-end aspect of Deep Learning. Giving birth to *MicroStateNet*. However, due to the nature of segmentation, there is a complete loss of spectral information and a partial loss of temporal and spatial features compared to EEG-based CNNs. This limitation was the main motivation for creating *DeepStateNet*. DSN was designed as a multimodal model, combining an existing EEG-based MI-BCI architecture with a microstate-based approach. This design makes it possible to capture both EEG and microstates features through parallel branches, which are then combined before the final classification layer, as illustrated in Figure 4.

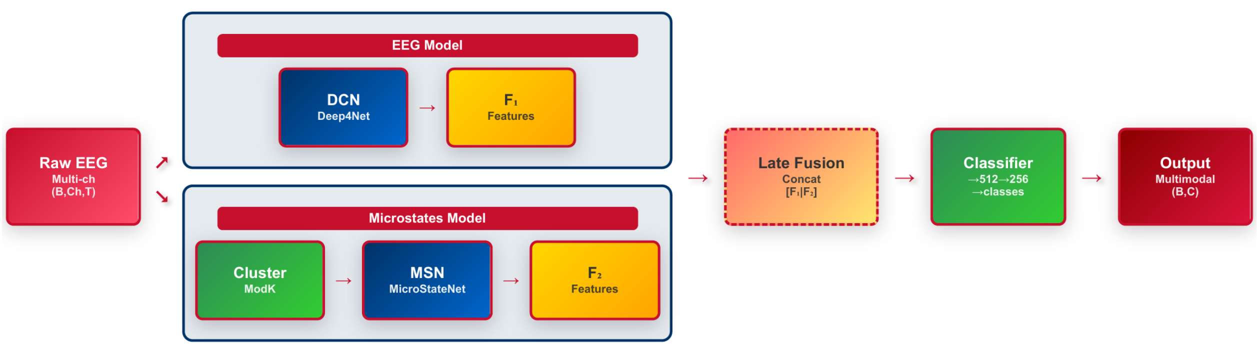

Figure 4: DeepStateNet Architecture

2.6 Hardware and Software

All models were implemented in *Python 3.11.9* using the *PyTorch* framework. Training was performed on either a *NVIDIA Tesla V100* or a *NVIDIA T400* GPU. The *ModKmeans* clustering algorithm was used from the *Pycrostates* package (v0.7.0). For compatibility with *PyTorch*, the *DeepConvNet* model was from the *Deep4Net* implementation provided in the *Braindecode* library (v0.8.1) [5]. All source code is available in the associated *GitHub repository*.

3 Results

3.1 Microstates

3.1.1 K-clustering

Figure 5 shows that both GEV and spatial correlation rise in a logarithmic manner as K increases, while the CH index decreased following an exponential decay. To balance these divergent tendencies, a joint selection criterion was applied. For GEV and spatial correlation, a plateau detection criterion was used: when the relative improvement between consecutive K values dropped below 1%, the metric was considered to have stabilized. For the CH index, the optimal K was estimated using the *Kneedle* algorithm [28], which detects inflection points corresponding to the "knee" of the curve. This combined strategy ensured that the final cluster solutions did not deviate to much from the theorized canonical value of five microstates [23], while avoiding overfitting from unnecessarily large K values.

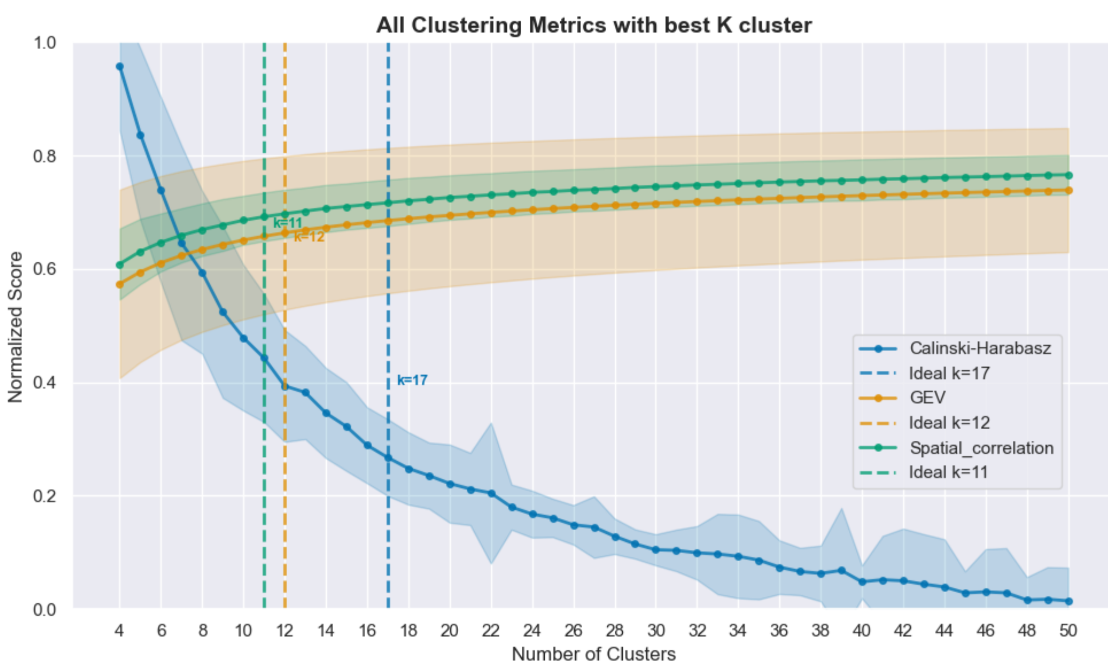

Figure 5: Normalize scores distribution from 4 to 50 clusters. The score are Global Explained Variance (GEV), Spatial Corelation and normalized Calinski-Harabasz (CH). For each score, the best k cluster is estimated using increment threshold of 1% for GEV and spatial correlation and a Knee

As illustrated in Fig. 5, the three criteria suggested slightly different optima: the spatial correlation reached a plateau at $K = 11$, the GEV at $K = 12$, and the Calinski–Harabasz index at $K = 17$. A compromise solution of $K = 12$ clusters was therefore adopted, as it maximized both GEV and spatial correlation while remaining reasonably close to the canonical reference of $K = 5$ microstates reported in the literature. The Calinski–Harabasz index at $K = 17$ was considered an informative indicator; however, limited confidence in the knee-point detection procedure led to a more conservative choice of $K = 12$. It should nevertheless be acknowledged that this decision is partly arbitrary, as it relies on heuristic thresholds (e.g., the 1% plateau rule) and algorithmic approximations, which may constitute a limitation of the present approach.

Table 3: Score summary for selected K clusters, where GEV Global Explained Variance, SC is Spatial Corelation and CH is normalized Calinski-Harabasz.

K	GEV Mean (%)	GEV Std (±%)	SC Mean (%)	SC Std (±%)	CH Mean (%)	CH Std (±%)
5	59.46	15.88	63.10	5.76	83.75	15.14
12	66.36	13.54	69.73	4.25	39.38	9.86

3.1.2 Microstates clusters

With the number of Microstates determined, the clustering models for K=5 and K=12 were created and fitted each subject's dataset using the *fit()* function. The clusters are then available for each clustering models as attributes. In Figure 6 and 7, it is noticeable that some subjects share similar microstates which indicates that some cannonical microstates exist as stated in the literature [9, 23, 24]. However, Although certain microstates may be shared across multiple subjects, they are not necessarily assigned the same labels, which complicates direct comparison. Furthermore, some topographies appear inconsistent or implausible, suggesting amplitude discrepancies either across channels within a subject or between subjects. Unfortunately, since the plotting function in Pycrostates is tightly embedded within the MNE library [29], it was not possible to fully access or adjust the underlying plotting parameters.

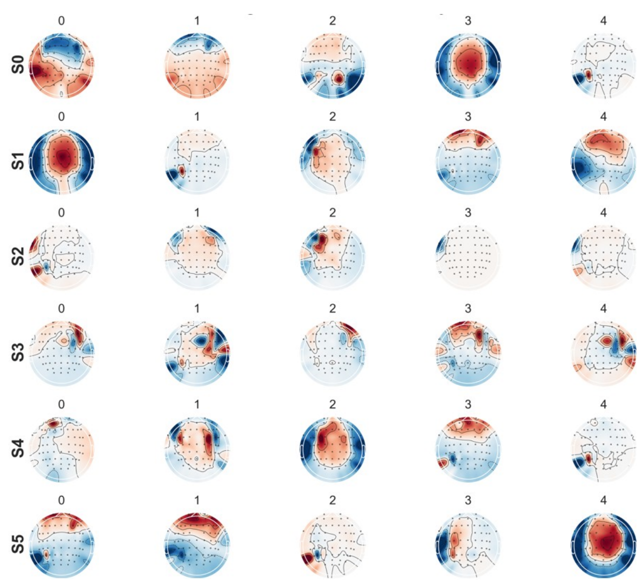

Figure 6: Pycrostates embedded MNE plot of EEG topography of Microstates for 5 first subject with K= 5. Subjects appear to share certain microstates, albeit with occasional mislabeling. However, some topology irregularities are evident, particularly subject 2 (S2) which displays extreme amplitude variations that differ markedly from the typical inter-channel or inter-subject patterns.

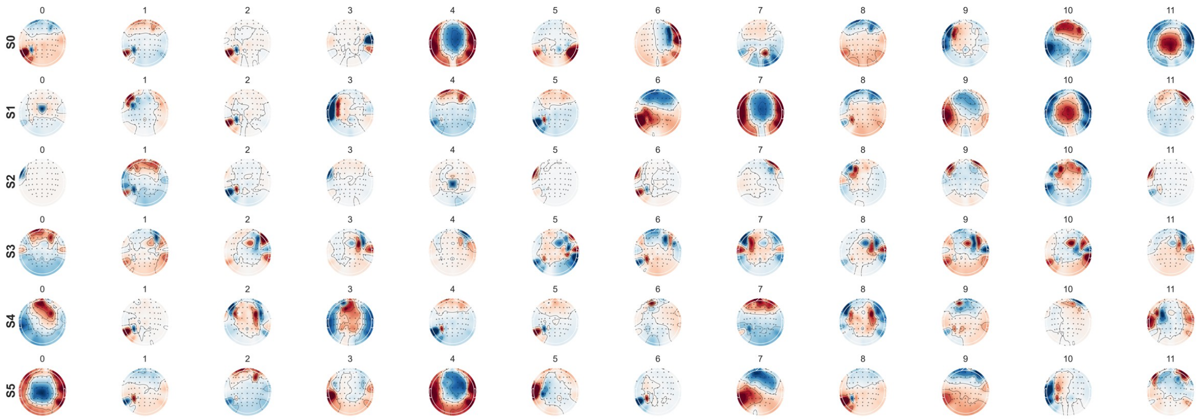

Figure 7: Pycrostates embedded MNE plot of EEG topography of Microstates for 5 first subject with K= 12. Subjects appear to share certain microstates, albeit with occasional mislabeling. However, some topology irregularities are evident, particularly subject 2 (S2) which displays extreme amplitude variations that differ markedly from the typical inter-channel or inter-subject patterns.

3.1.3 Microstates sequences

With the ModKmeans created, the microstates timeseries can be created using ModKmeans *predict()* function which uses spatial correlation to assign at each time t a microstates label to which it correlates the best.

As observed in Figure 8, each subject have predominant microstates but they are not the same between subjects. Moreover, a difference within subject's trials class is noticeable too. There seems to be a difference between close and open (row 2 and 4) in sequences and microstates predominance. Additionally, a difference is also noticeable between *rest* trials obtain from *open* and *rest* trials obtain from *close* trials. Indeed, they seem to be more similar between same trials dataset than same between *rest* trials.

Furthermore, in order to add another dimension of information, the corresponding normalized GFPs of each trial was concatenated to its corresponding microstates sequences, as they are intrinsically linked.

Table 4: Microstate Sequence Summary Statistics (N=50 subjects)

	5 Clusters			12 Clusters		
	Count	Duration (ms)	Proportion	Count	Duration (ms)	Proportion
Mean	61,512.8	40.2	0.200	26,982.5	32.9	0.083
Std Dev	8,391.2	13.5	0.113	2,530.1	3.9	0.056
Min	139.0	1.9	0.0005	93.0	16.9	0.0003
Max	147,619.0	132.9	0.757	104,953.0	118.1	0.378
Range	147,480.0	131.0	0.756	104,860.0	101.2	0.377

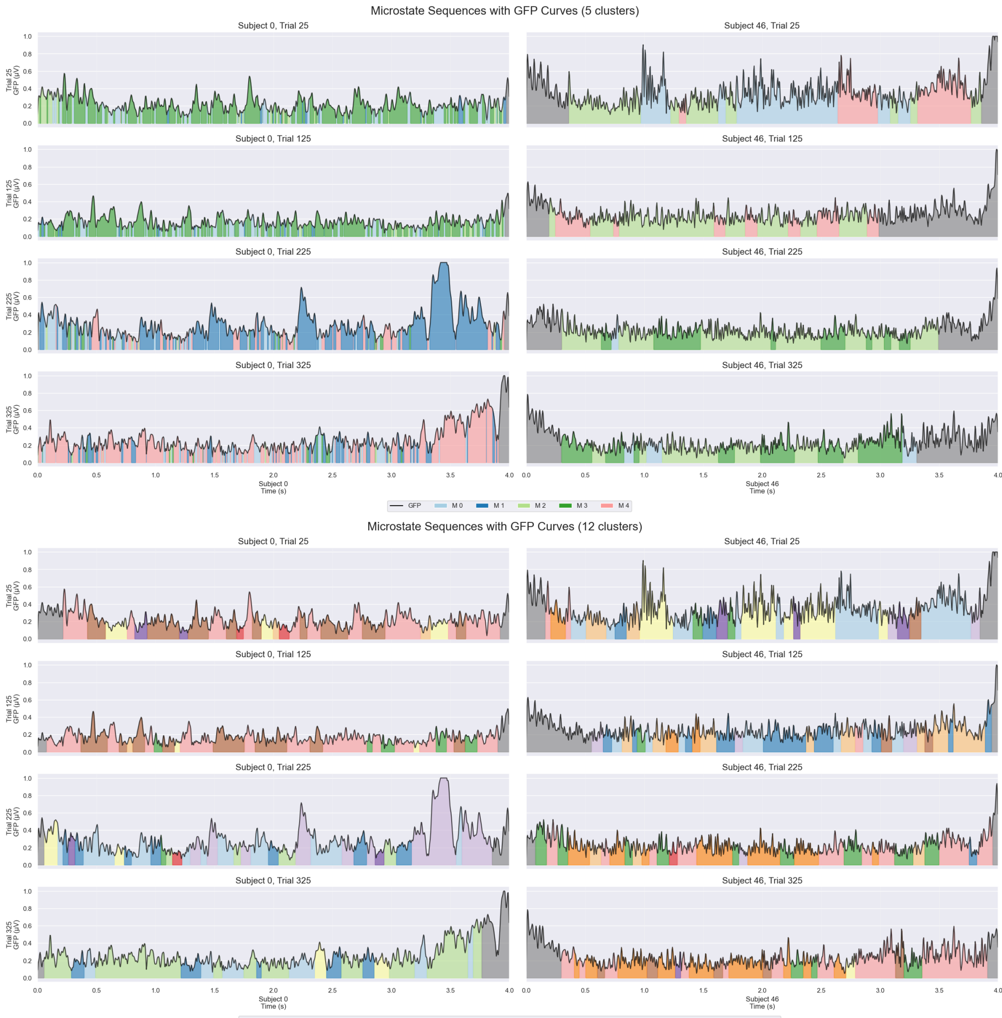

Figure 8: Microstates timeseries, segmentation plots for $k = 5$ and $k = 12$ clusters, where GFP is shown and under it's curve is the respective microstate segment assign to it. Results taken from two subjects (left:0, right:46). Each subject shows four trials, one for each class, from top to bottom: rest (from close trials) close, rest(from open trials), open trials. This Figure shows the difference in microstates predominance between subjects and within subjects. In grey are unlabeled segments which could not fit with any clusters

3.2 Models

3.2.1 Overall results

With all the microstates timeseries available for each subject and each K cluster (C5 and C12), the models are ready to be trained. Table 8 presents the mean score and standard deviation for each model and both scores (Balanced Accuracy top row and F1 Macro bottom row). Adding Figure 9's visuals, three groups of models are clearly distinguishable. In red, the baseline, DeepConvNet with an average Balanced accuracy of 73.8% and 68.6% F1. As for the second group (purple to yellow), with a lower than baseline average, all the MicrostatesNet variants models are averaging from $52.1 - 60.0\%$ bal. acc. and $44.1 - 56.8\%$

F1. For the last group (Blues and Greens), with a much higher scores from baseline are the DeepStateNet variants with a bal. acc. from $84.3 - 86.2\%$ and $84.0 - 85.9\%$ F1.

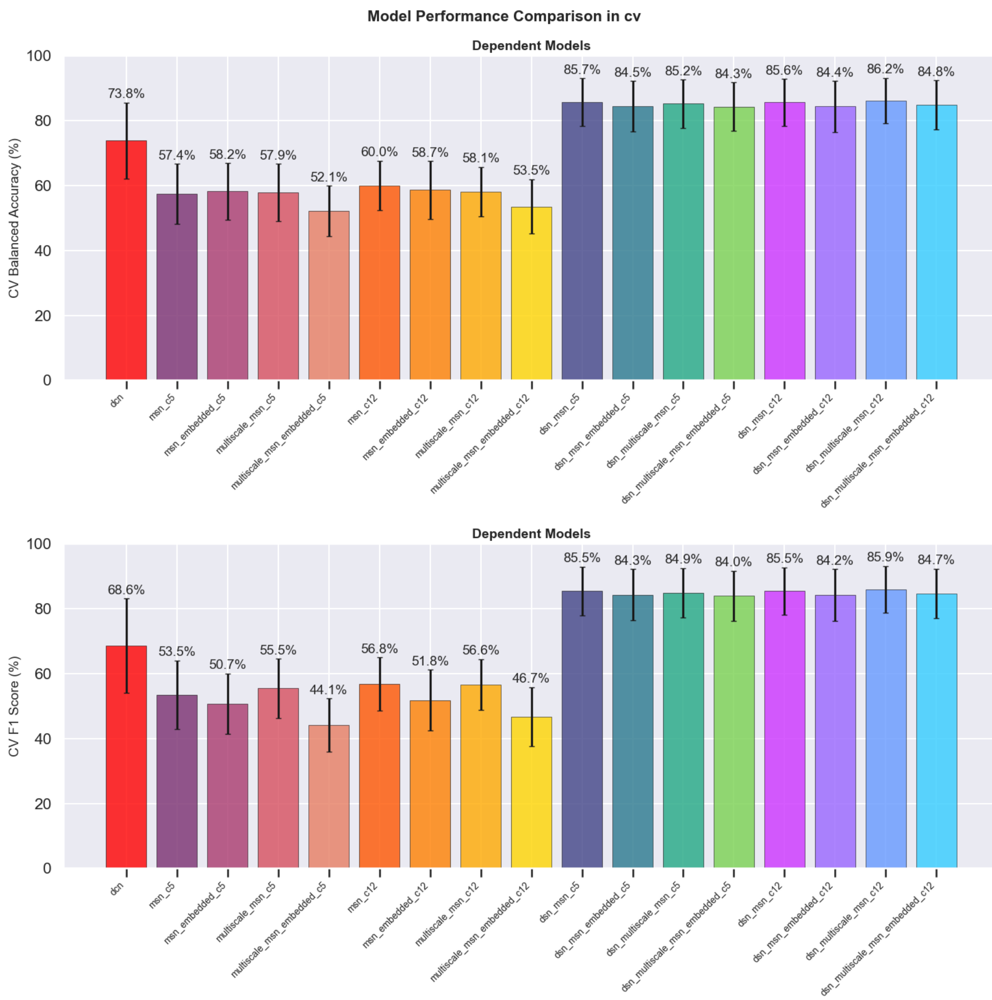

Figure 9: Bar plot of mean cross-validation across all 50 subjects for each model with standard deviation bar. In order, from left to right, there is DeepConvNet (DCN), MicroStateNet (MSN) variants and DeepStateNet (DSN) variants.

Figure 10 shows the same patterns for the medians of the scores as seen for the means. Additionally, the plot adds information about the distribution of each model across all subjects. It can be noticed that DeepConvNet (in red), has a larger density compared to the other two groups. Then, the MicroStateNet's group has a slightly tighter density and some intra-variability. As for the third group, DeepStateNet, a much more compact distribution can be observed for each and there are less differences between each variant.

3.2.2 Exploring model parameters

With these results, the best parameters for MicroStateNet and DeepStateNet can be determined. In order to do this, a exploratory comparison was done for each parameter:

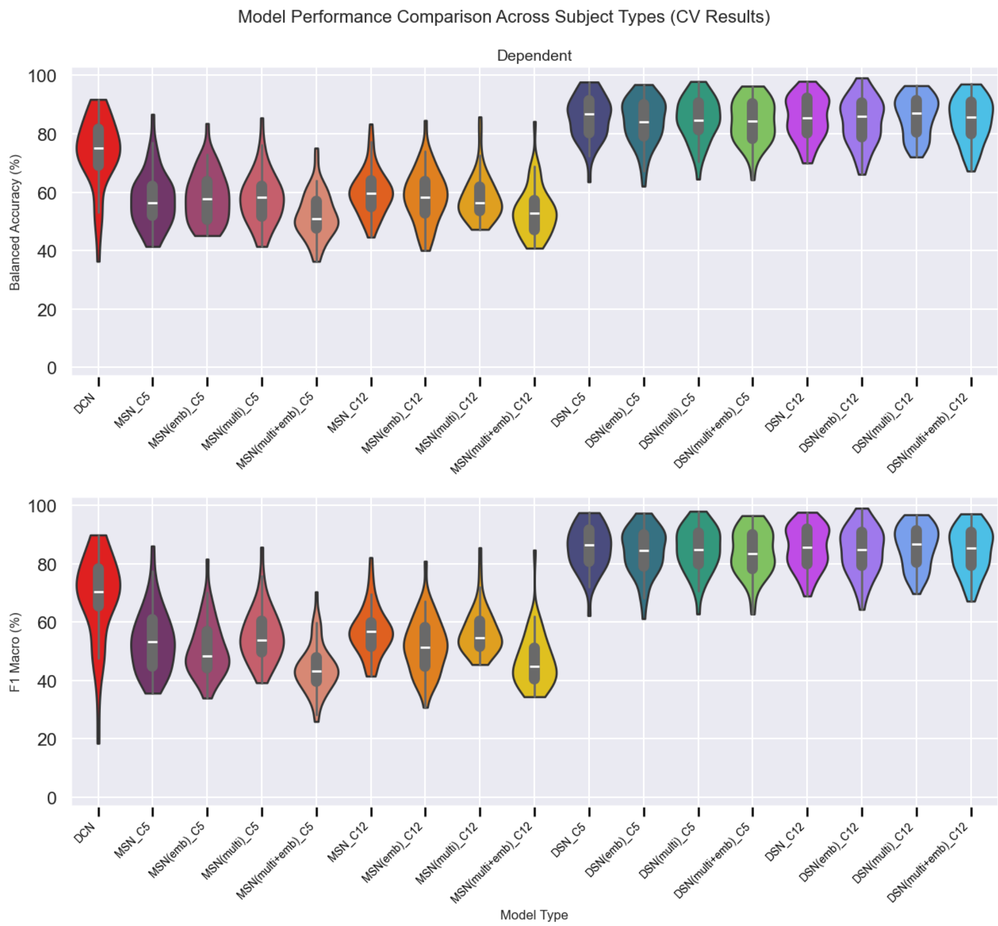

Figure 10: Violin plot of mean cross-validation across all 50 subjects for each models with inner box-plot. In order, from left to right, there is DeepConvNet (DCN), MicroStateNet (MSN) variants and DeepStateNet (DSN) variants.

- number of clusters (5 vs 12 microstates)
- scale (1 vs multi kernel size)
- encoding (one-hot vs embedding)

A complete ANOVA analysis followed by a Pairwise comparison with Benjamini–Hochberg false discovery rate (*fdr_bh*) for multiple test correction follow by a pairwise comparison as post-hoc test was done for each parameter. Table 5 does not show any significant difference for any group after correction. This indicates that there are no overall significant differences between five and twelve clusters, nor there is between having one kernel size and three parallel branches, as well as between one-hot encoding and embedding when looking at the balanced accuracy. Nevertheless, by looking into Table 6, a significant result is noticeable for encoding group. Making it interesting to look more into it and see within the groups.
Concerning the post-hoc pairwise comparison it was done with a corrected pairwise t-test

Table 5: One-way ANOVA Results for balanced accuracy score Model Performance Comparisons. Showing results for all three parameters to find the final parameters for MicroStateNet and DeepStateNet.

Comparison	Source	Sum of Squares	df	Mean Square	F	p-value
Number of Clusters	Condition	469.73	1	234.87	0.91	0.402
	Residual	218,098.64	847	257.40	–	–
	Total	218,568.37	849	–	–	–
Scale	Condition	658.25	1	658.25	2.56	0.110
	Residual	217,910.11	848	257.00	–	–
	Total	218,568.36	849	–	–	–
Encoding	Condition	979.87	1	979.87	3.82	0.051
	Residual	217,588.49	848	256.62	–	–
	Total	218,568.36	849	–	–	–

Note: None of the comparisons reached statistical significance at $\alpha = 0.05$. *Encoding shows marginal significance (p = 0.051).*

Table 6: One-way ANOVA Results for F1 macro Score Model Performance Comparisons. Showing results for all three parameters to find the final parameters for MicroStateNet and DeepStateNet.

Comparison	Source	Sum of Squares	df	Mean Square	F	p-value
Number of Clusters	Condition	300.32	1	150.16	0.44	0.645
	Residual	290,345.28	847	342.91	–	–
	Total	290,645.60	849	–	–	–
Scale	Condition	294.78	1	294.78	0.86	0.354
	Residual	290,350.82	848	342.57	–	–
	Total	290,645.60	849	–	–	–
Encoding	Condition	3,359.14	1	3,359.14	9.92	0.002**
	Residual	287,286.46	848	338.79	–	–
	Total	290,645.60	849	–	–	–

Note: Only the encoding comparison was statistically significant at $\alpha = 0.05$. *** indicates* $p < 0.01$.

and computing *cohen's d* and its respective confidence interval:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p} \tag{5}$$

where the pooled standard deviation is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{6}$$

The confidence interval for Cohen's d is calculated as:

$$CI = d \pm t_{\alpha/2,df} \cdot SE_d \tag{7}$$

where the standard error is:

$$SE_d = \sqrt{\frac{n_1 + n_2}{n_1 \cdot n_2} + \frac{d^2}{2(n_1 + n_2)}} \tag{8}$$

and where $t_{\frac{\alpha}{2},df}$ is the critical t-value for $\frac{\alpha}{2}$ with $df = n_1 + n_2 - 2$, $\bar{x}_1$ and $\bar{x}_2$ are the sample means, s_1^2 and s_2^2 are the sample variances, and n_1 and n_2 are the sample sizes for groups 1 and 2, respectively.
Looking at the pairwise comparison presented in the forest plots in Figure 11, it is indeed noticeable that no models differs between the two groups of clusters, even though the *Cohen's d* effect tends to twelve cluster due to one variant which is MSN.

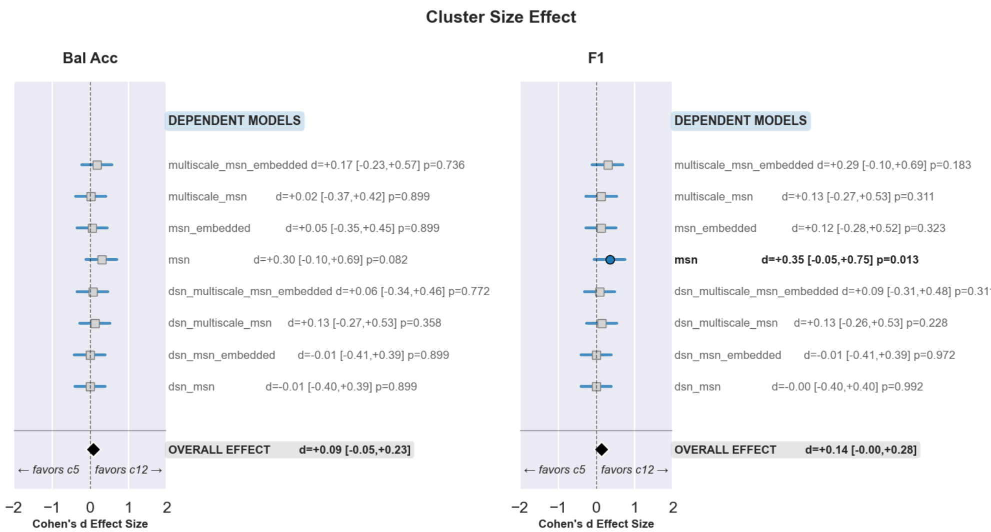

Figure 11: Forest plot for number of cluster as criteria, between 5 or 12 clusters

Figure 12 present two to three paires with individually significant difference and two of them with large enough effect. Although this concern only MicroStateNet variants. There is also a slight incline, yet without enough effect, for a multiscale model. Therefore, there is no overall significant differences between scaling parameters.
In the case of encoding effect, a much larger effect can be observed in Figure 13. This is especially true for F1 score (right) where all pairs are significantly different and three of them yield enough effects. This explains the previously presented ANOVA results. Although not all pairs are significant in balanced accuracy, the overall indicates a preference for embedding. For the the final evaluation, only one model variant must be chosen, which is why the following parameters were selected as the final MicroStateNet and DeepStateNet:

- Number of clusters: 12
- Scaling: Multiscale
- Encoding: Embedding

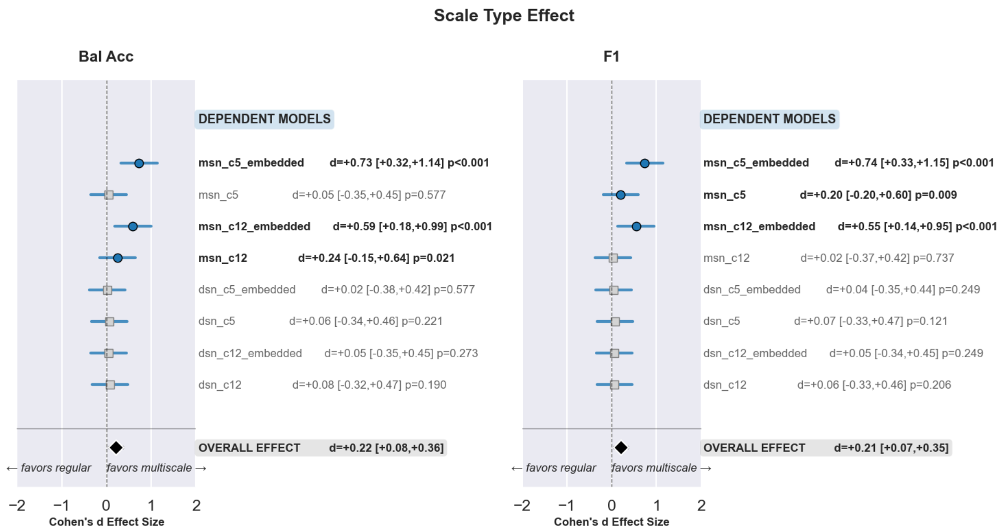

Figure 12: Forest plot for number of kernel size as criteria, between 1 and 3 different kernel size in parallel.

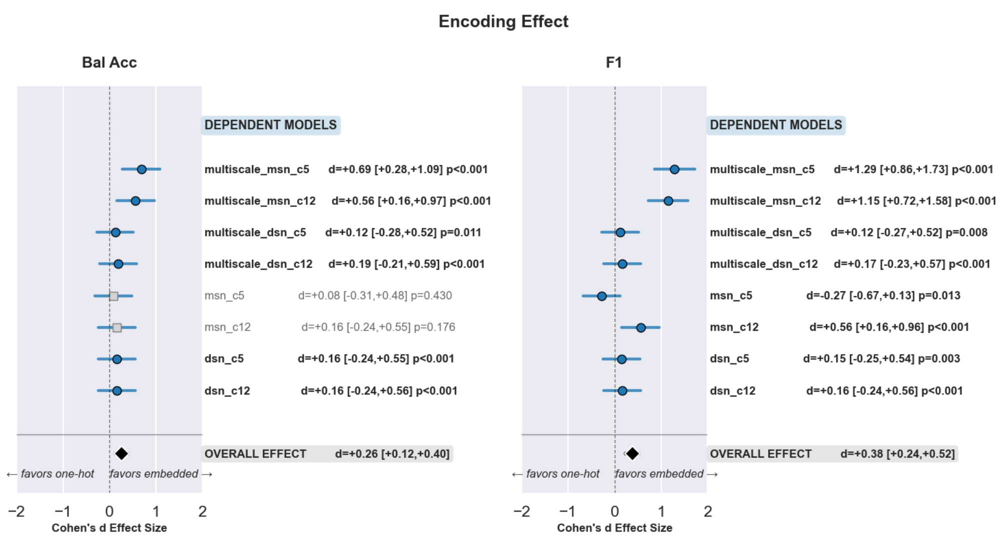

Figure 13: Forest plot for encoding type as criteria, one-hot encoding and embedding.

3.3 Final comparison

For the final comparison, all model types (MSN and DSN) with the previously selected parameters will also be compared using an ANOVA with FDR_H corrections and a pairwise comparison for post-Hoc analysis.
In both tables 9a and 10a, there is a clear significant difference in both balanced accuracy and F1 macro.
Table 9b and 10b show clearly which model perform better. The lowest to highest models for both scores is MicroStateNet, then DeepConvNet and finally DeepStateNet. Figure 14 illustrate it better, where the boxplot shows a clear difference between each model with all p-values < 0.001 and each with large $Cohen'sd$ indicating effect size, all $|d| > 1$ as shown in

Table 7a and 7b. This indicates that MicroStateNet alone is not enough, this Deep Learning paradigm for microstates feature is better than chance level (three classes: $> 33.\overline{3}\%$) but still lacking in comparison with existing model such as DeepConvNet. However, combining the two models and encapsulating both the EEG and microstates features appears to significantly improve the classification.

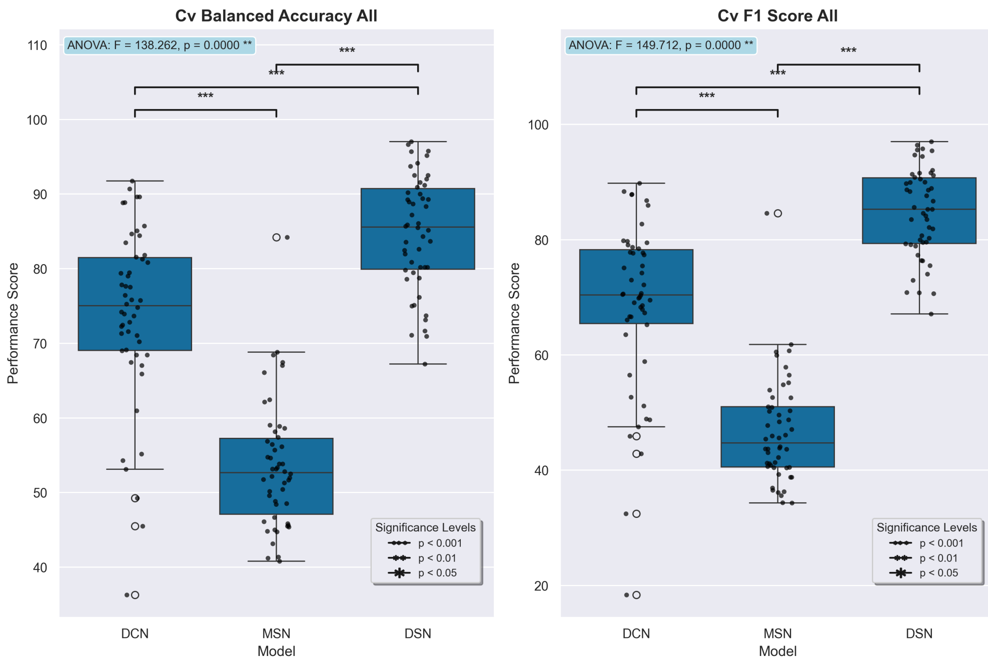

Figure 14: Boxplot distributions of model performance for DeepConvNet (DCN), MicroStateNet (MSN), and DeepStateNet (DSN) using the selected parameters. Results of the ANOVA analysis are reported together with significance level indicators, with *p-values* corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (*fdr_bh*) procedure. The left plot shows Balanced Accuracy scores, while the right plot shows F1-macro scores. In both metrics, significant differences were observed between all models, with performance ranking MSN < DCN < DSN.

4 Discussion

4.1 Results analysis

4.1.1 K clustering

The clustering for both choices of cluster number yielded a GEV of approximately 60%, which is close to the literature average reported for the four to five principal clusters [10]. Although slightly lower, these values remain reasonable and can be considered satisfactory. This outcome reflects a compromise between GEV spatial correlation and the Calinski-Harabasz index, even though the final decision on k was ultimately arbitrary and could benefit from more principled selection methods.

Table 7: Post-hoc Pairwise Comparisons for DCN vs MSN vs DSN Models

(a) Balanced Accuracy Post-hoc Results

Comparison	Mean 1 (%)	Mean 2 (%)	Difference	Cohen's d	t-statistic	p-value
DCN vs MSN	73.82	53.54	20.28	1.96	10.08	$< 0.001 ***$
DCN vs DSN	73.82	84.79	-10.97	-1.09	-7.51	$< 0.001 ***$
MSN vs DSN	53.54	84.79	-31.25	-3.87	-24.64	$< 0.001 ***$

(b) F1 Score Post-hoc Results

Comparison	Mean 1 (%)	Mean 2 (%)	Difference	Cohen's d	t-statistic	p-value
DCN vs MSN	68.61	46.73	21.88	1.78	8.63	$< 0.001 ***$
DCN vs DSN	68.61	84.65	-16.04	-1.36	-7.98	$< 0.001 ***$
MSN vs DSN	46.73	84.65	-37.92	-4.45	-26.52	$< 0.001 ***$

*Note: *** indicates $p < 0.001$ after Bonferroni correction. All pairwise comparisons are statistically significant. Effect sizes: small ($0.2 \leqslant |d| < 0.5$), medium ($0.5 \leqslant |d| < 0.8$), large ($|d| \geqslant 0.8$). DSN significantly outperforms both DCN and MSN, while DCN significantly outperforms MSN.*

When examining the clusters in Figure 6 and Figure 7, some redundancy can be observed across subjects. At the same time, several clusters resemble those reported in previous studies [21, 23], which indicates that the clustering remains reasonably consistent with established microstates topographies. However, discrepancies are also present, which may arise from differences in EEG amplitude between subjects or from electrode-specific variations within the same subject, as illustrated for subject 2 depending on the plotting function of *Pycrostates* [26].

4.1.2 Segmentation

Regarding segmentation, Figure 8 highlights both inter- and intra-subject variability. Although each subject was segmented into the same number of clusters, the predominant microstate labels differed, suggesting either subject-specific cluster solutions or possible mislabeling. Furthermore, variability was also observed across trials within a single subject. As seen in the same figure, while the same microstates are present across *open*, *close*, and *rest* trials, their relative predominance differs. Interestingly, *rest* trials do not cluster together; instead, they appear more similar to the action block from which they were extracted. For instance, *rest* segments from the *close* block tended to share the same predominant clusters as *close* trials, with a similar effect observed for *rest* from the *open* block.

Finally, Table 4 shows that the mean duration of microstates is around 30–40 ms, reaching up to 130 ms, which accords with previously reported findings [21, 23].

4.1.3 Models

The overall results can be grouped into three main categories: DCN, MSN variants, and DSN variants expressing a clear distinction between each model type. While the differences are observed between groups, there is little variation within groups. The exploration of MicroStateNet design revealed no substantial or consistent improvements across parameter variants, as neither significant differences nor strong effect sizes were found. The only exception concerns the encoding strategy, where embedding showed a significant improvement in F1 score and a larger but still moderate effect size. This suggests that embedding is a promising choice, but its potential may require a more refined implementation to be fully realized.

4.1.4 Final results

Ultimately, the final results indicate a clear difference between MicroStateNet, DeepConvNet, and the proposed DeepStateNet. MicroStateNet achieves only a modest classification accuracy above chance level, although lower than the baseline DeepConvNet. This suggests that relying only on microstates features is insufficient, as a substantial amount of information is lost compared to models trained directly on EEG signals. In contrast, DeepStateNet shows a clear improvement over DeepConvNet, suggesting that the fusion of EEG and microstate features provides complementary information and leads to enhanced performance.

This confirms the main motivation of this work. While microstates alone are not sufficient for accurate classification, their integration with EEG signals provides complementary and neurologically meaningful information. In this sense, the results support the idea that introducing microstate knowledge into deep learning models at the cost of breaking the end-to-end paradigm can nonetheless help the model and improve performance.

4.2 Limitations

4.2.1 Dataset

The dataset used in this study originates from a previously conducted experiment specifically designed for MI-BCI EEG research [3]. The data was provided already preprocessed according to standard procedures for EEG analysis.
Additionally, it should be noted that during the experiment many patients were unable to fully control their hand movements. Consequently, EMG artifacts were present in several trials, particularly during the close-hand condition. Although these artifacts were addressed during preprocessing, some residual contamination may remain in a few subjects.

4.2.2 ModKmeans and K selections

A few limitations should be considered regarding the clustering procedure. The selection of the number of clusters (k) relied on a combination of plateau detection for GEV and spatial correlation as well as a knee-point detection for the Calinski-Harabasz index. Both the threshold for plateau and the resulting choice of k remain somewhat arbitrary, reflecting the inherent subjectivity in such criteria. More generally, Kmeans clustering itself has limitations, as it depends on a spatial similarity metric to define clusters. Alternative metrics, such as Global Map Dissimilarity (GMD), have been used in other studies [9, 12]. In addition, the algorithm initializes clusters randomly multiple times, which can lead to convergence on local maxima, potentially affecting the final cluster solution.

Finally, it should be noted that processing EEG signals into microstates sequences prior to feature extraction partially breaks the end-to-end principle of deep learning, as the model cannot learn representations directly from the raw EEG signals but instead operates on clustered microstates sequences.

4.2.3 Models

As this work was exploratory, no hyperparameter tuning was performed. Most parameters were chosen based on previously established EEG models such as DeepConvNet [5]. This is the case for the selection of kernel sizes, which may not capture all relevant features and could benefit from alternative or additional sizes. Furthermore, for the multi-scale architecture, the number of layers of the convolution blocks was reduced from four to three and only three kernel size were use for computational feasibility, which may have constrained the model's representational capacity. Finally, the current method has a limited global view of microstates features across trials or subjects, which may restrict its ability to fully exploit the temporal patterns inherent in microstates dynamics.

4.3 Future Prospects

4.3.1 Clustering

Future improvements could focus on refining the microstates clustering procedure. Specifically, methods for selecting the optimal number of clusters (k) could be developed that avoid arbitrary threshold decisions, providing a more objective criterion for cluster determination. Similarly, the choice of distance or similarity metrics, which currently affects the definition of clusters, could be further investigated.
Another promising direction would be to integrate clustering directly within a deep learning framework, thereby preserving the end-to-end learning principle. For instance, a deep clustering model could be jointly trained with DeepStateNet or other multimodal networks, allowing the latent representations of EEG signals to inform cluster formation. Projecting EEG data into a latent space prior to clustering may also provide a more robust metric for defining similarity between microstates, potentially improving both the interpretability and generalizability of the resulting clusters.

4.3.2 Improving Models

Another option could be to focus on improving the neural network architecture and training strategies. Hyperparameter exploration, including variations in kernel sizes and the depth of convolutional layers, may result in better feature extraction from microstate sequences. In addition, the integration of attention mechanisms or transformer-based layers could provide a parallel pathway to capture global patterns across time, similar to the approach employed in CASTNet [3].

4.3.3 Other case study

Other than improving the models, MicroStateNet and DeepStateNet could also improve using other case studies, such as independent subject and adaptive subject models [6]. Independent subject would be a *Leave-One-Subject-Out* (LOSO), which would be training the model on all subject excluding the one we design the model for, then validating the model on this particular subject's data. This means having a model with no prior information on the target subject before validation.

This would allow to take into account the variance between individuals and increase the training dataset. However, this could at first have worst results for the same reason. Nonetheless, adaptive subject could solve this problem. Because adaptive subject is, like independent, doing a LOSO on all other subjects, but additionally fine-tuning the model on the subject of interest. This might be a way to combine subject dependent and independent case study in order to have an enhanced training set expressing inter-subject variability while fitting a model for the specific subject.

While this case study looks promising, some considerations must be taken into account. First, expanding the dataset to include all subjects greatly increases memory requirements and training time, necessitating more powerful hardware or substantially more RAM and processing time. Moreover, given the nature of microstates, there is variability across subjects, as well as inconsistent labeling of some similar microstates (see Figure 8). To address this, it is essential to establish a method for linking microstates across subjects. One possible method would be to train a ModKmeans model jointly on all subjects, though this would demand considerable RAM resources. An alternative approach would be to reduce the number of distinct microstates in the already extracted sequences by relabeling and merging similar patterns to ensure consistency across subjects. Particularly because the theoretical maximum number of unique microstates grows as $n_{subjects} \times k_{microstates}$.

Beyond model refinements, additional studies and larger datasets could be useful, not to redesign the current study, but to expand the data and obtain further evidence to support the underlying theory.

5 Conclusion

In conclusion, the objective of this exploratory research was to investigate how an existing MI-BCI model could be enhanced with a neuroscience perspective, specifically through the microstates paradigm, even at the risk of breaking the conventional *end-to-end* approach of Deep Learning. Throughout this project, the main focus was first to examine the principal shapes of microstates and to determine how EEG data should be processed in order to correctly extract microstate features. Decisions were guided by the state of the art in both EEG-based MI-BCI and microstates clustering. This was followed by the design of a CNN architecture adapted to microstate feature extraction, and finally by the exploration of multimodal strategies to combine EEG and microstate features. DeepStateNet presented significant improvements, suggesting that incorporating microstate features can positively contribute to MI-BCI performance, and thus represent a promising direction for further research.

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6 Annexes

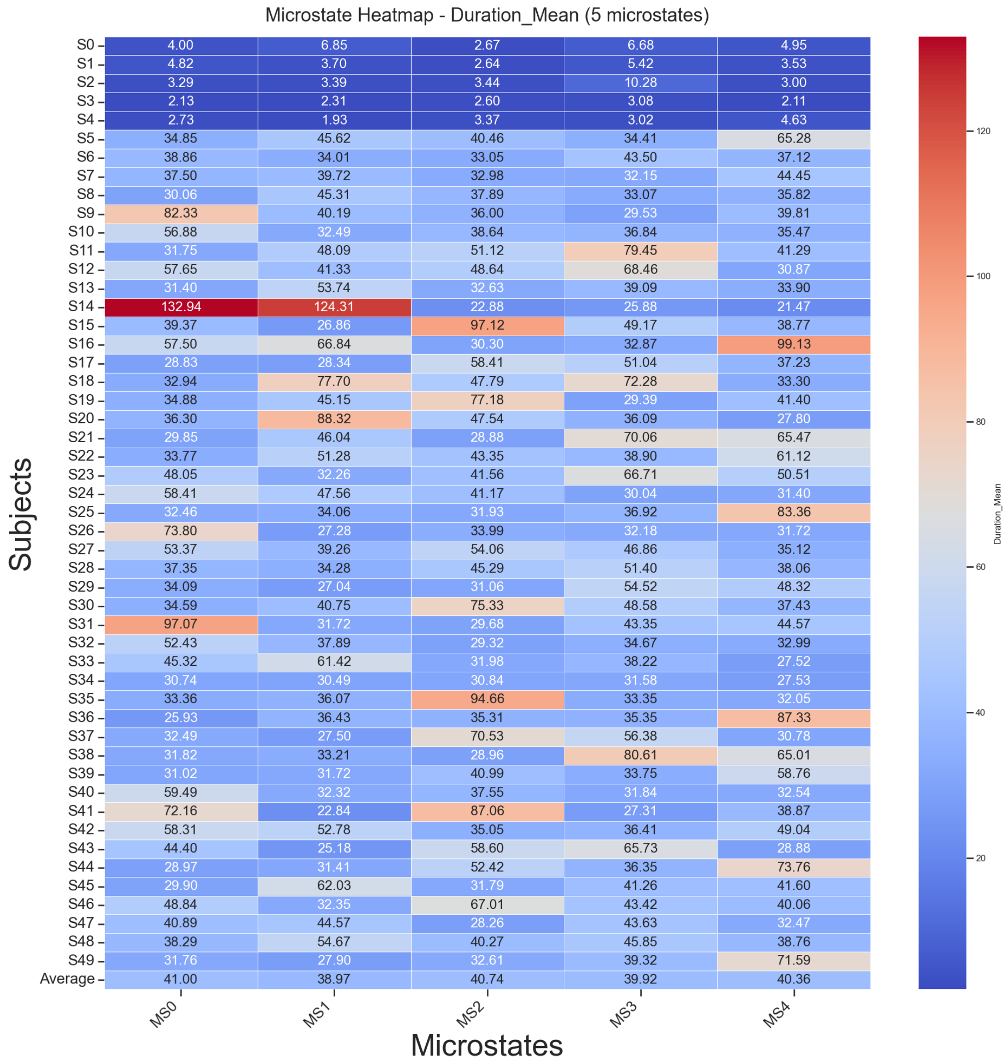

Figure 15: Heatmap for mean duration of each microstates in each subjects (c=5)

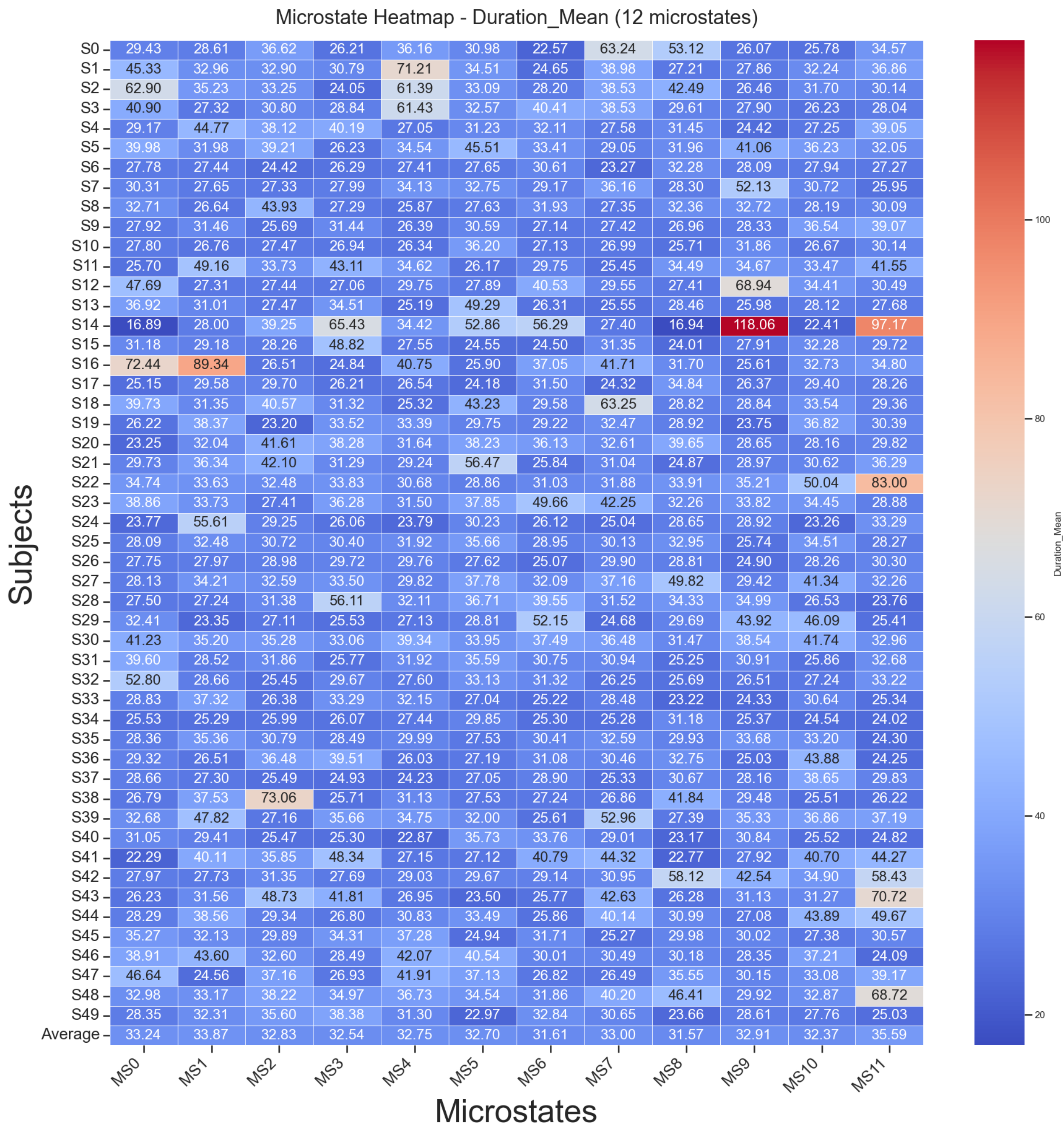

Figure 16: Heatmap for mean duration of each microstates in each subjects (c=12)

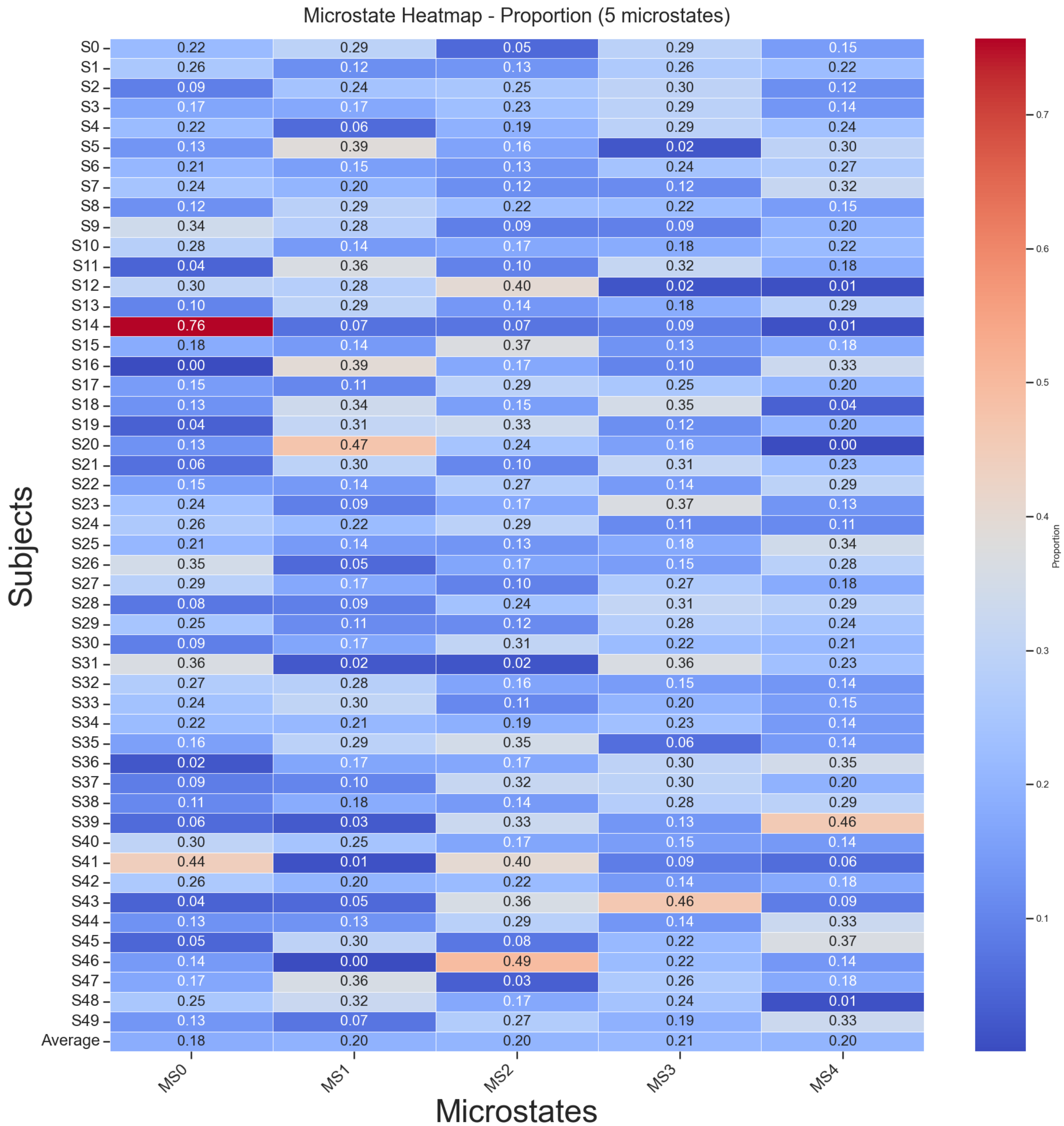

Figure 17: Heatmap for proportion of each microstates in each subjects (c=5)

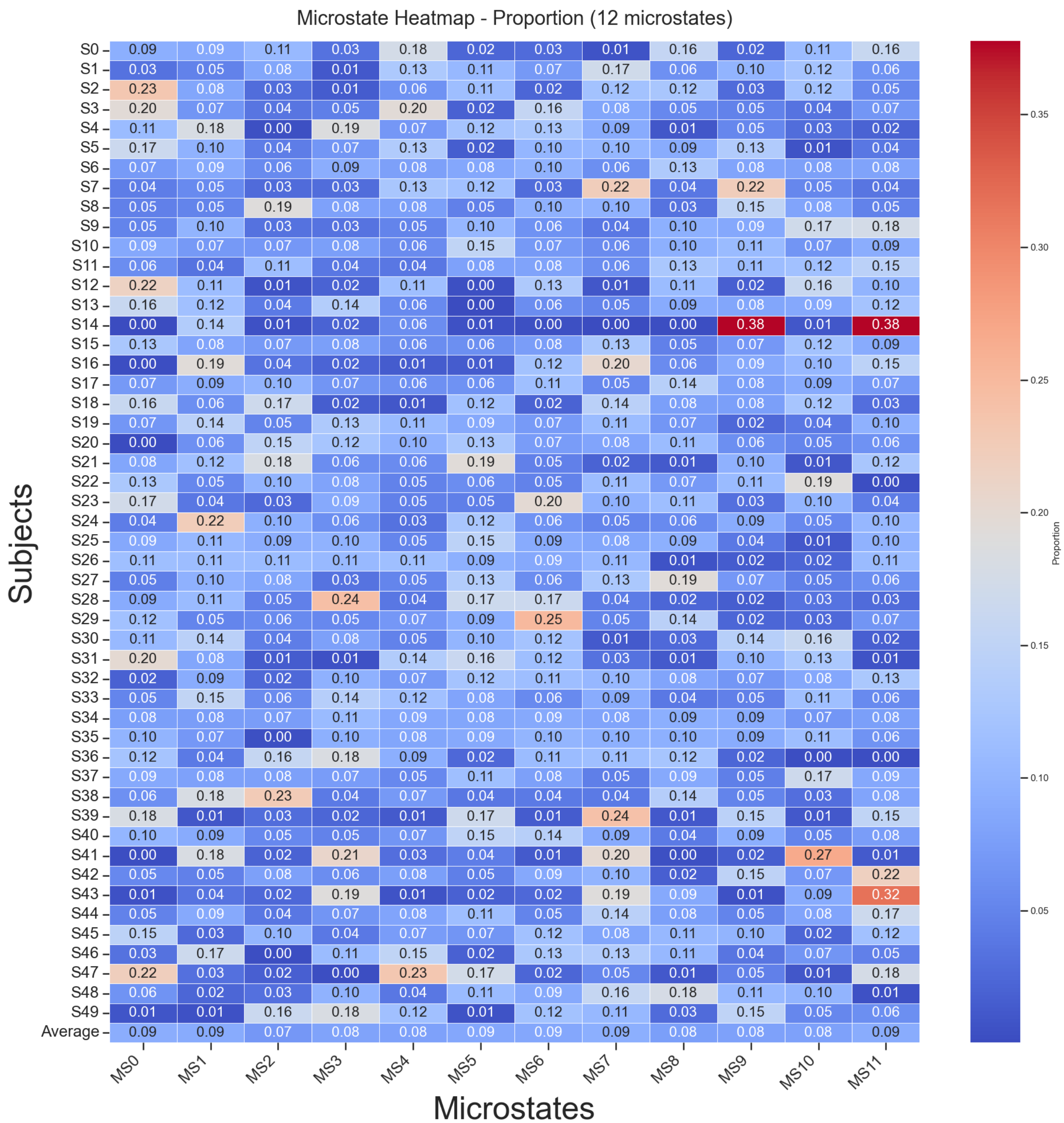

Figure 18: Heatmap for proportion of each microstates in each subjects (c=12)

Table 8: Model Performance Summary

Model Name	Balanced Accuracy		F1 Score	
	Mean (%)	Std	Mean (%)	Std
DCN model				
dcn	73.82	11.80	68.61	14.61
MSN models				
msn_c5	57.45	9.28	53.46	10.62
msn_c12	59.99	7.66	56.81	8.19
msn_embedded_c5	58.22	8.73	50.70	9.37
msn_embedded_c12	58.68	8.93	51.83	9.38
multiscale_msn_c5	57.90	8.85	55.49	9.13
multiscale_msn_c12	58.10	7.68	56.61	7.86
multiscale_msn_embedded_c5	52.14	7.77	44.14	8.24
multiscale_msn_embedded_c12	53.54	8.39	46.73	9.13
DSN models				
dsn_msn_c5	85.69	7.36	85.46	7.54
dsn_msn_c12	85.63	7.22	85.45	7.28
dsn_msn_embedded_c5	84.48	7.78	84.32	7.86
dsn_msn_embedded_c12	84.39	7.87	84.23	7.97
dsn_multiscale_msn_c5	85.23	7.44	84.93	7.64
dsn_multiscale_msn_c12	86.18	6.98	85.93	7.22
dsn_multiscale_msn_embedded_c5	84.33	7.42	83.97	7.77
dsn_multiscale_msn_embedded_c12	84.79	7.60	84.65	7.68

Table 9: One-way ANOVA Results for DCN vs MSN vs DSN Model Comparison (Balanced Accuracy)

(a) ANOVA Summary

Test	F-statistic	p-value	Significant
Model Type	138.26	$< 0.001***$	Yes

(b) Descriptive Statistics by Model Type

Model	number of subjects	Mean (%)	Std Dev	SEM
DCN	50	73.82	11.92	1.69
MSN	50	53.54	8.47	1.20
DSN	50	84.79	7.67	1.09

*Note: *** indicates $p < 0.001$. DSN shows highest performance, followed by DCN, then MSN.*

Table 10: One-way ANOVA Results for DCN vs MSN vs DSN Model Comparison (F1 Score)

(a) ANOVA Summary

Test	F-statistic	p-value	Significant
Model Type	149.71	$< 0.001***$	Yes

(b) Descriptive Statistics by Model Type

Model	number of subjects	Mean (%)	Std Dev	SEM
DCN	50	68.61	14.76	2.09
MSN	50	46.73	9.23	1.30
DSN	50	84.65	7.76	1.10

*Note: *** indicates $p < 0.001$. DSN shows highest F1 performance, followed by DCN, then MSN.*